Supplementary material

AdaDINO-LA: A Controlled Study of DINOv2 Adaptation Strategies for Left Atrial Segmentation in LGE-MRI

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Tables and figures numbered with the prefix S are in this file; all other table, figure, and section numbers refer to the main article.

Supplementary methods

Hierarchical decoder

The hierarchical decoder of the decoder-capacity analysis (Section 4.4) follows the UNETR design. Tokens from encoder blocks 23, 17, 11, and 5 are each projected to 256 channels and enter a top-down path at 32, 64, 128, and 224 pixels, where they are concatenated with the upsampled path and fused by two convolutional blocks; a convolutional stem on the input slice adds a skip connection at 448 pixels. The decoder has 3.12 million parameters, against 0.66 million for the lightweight decoder with the four-layer readout. Subsets, batch order, learning-rate rule, checkpoint rule, and evaluation equal those of the common-recipe and label-budget runs, which serve as lightweight-decoder counterparts. The planned contrasts were hierarchical minus lightweight decoder, nnU-Net minus hierarchical decoder, and the ranking of strategies.

Training, reference, and external-data details

The first runs of four DINOv2 configurations computed the loss under autocast; they were retrained for all seeds, and only the retrained runs are reported. The nnU-Net reference used batch size two, Dice plus cross-entropy with deep supervision, SGD with Nesterov momentum and polynomial decay, checkpoint selection by a moving average of validation-patch Dice, and sliding-window inference with mirroring; the 2D configuration used the same plans file. In the label-budget analysis, a 50% budget was dropped for compute time before any run; all label-budget runs used one RTX 5090, and a rerun of one 108-volume LoRA run reproduced its Dice within 0.11 points. The CDEMRIS patients excluded a priori because their volumes carry Utah-style headers are p6, p8, p13, p15, p16, p20, p22, p24, p27, and p30.

Statistical details

For every contrast the random-number generator is reset to seed 20260926, and two-sided p values are twice the smaller bootstrap tail probability. The 16-contrast family of Table 4 includes three nnU-Net reference contrasts. The case $\times$ seed bootstrap of the key contrasts (Table S4) resamples cases and, independently for each configuration, the three seeds, so its intervals approximately reflect retraining variation; with three seeds this approximation is coarse. For the label-budget and decoder contrasts, whose configurations share one subset per seed, the three seeds are resampled jointly across configurations, so the intervals also reflect subset variation. The least-similar subset contains the 21 test volumes (two thirds) with the lowest maximum image correlation to any training volume in the image-similarity audit. The robustness criterion (an interval excluding zero and the same sign in all three per-seed differences) is a minimum screen rather than a test.

Table S1. Parameter counts and completed training lengths under the main protocol. Epoch triplets follow seed order 42 / 2026 / 3407. Total counts include the encoder, decoder and adapters where applicable.

Backbone	Adaptation	Total parameters	Trainable (%)	Epochs
Large	Frozen	305,023,969	0.215	31 / 26 / 34
Large	Partial FT	305,023,969	16.736	25 / 32 / 16
Large	LoRA	305,613,793	0.407	25 / 50 / 22
Large	Full FT	305,023,969	100.000	18 / 28 / 21
Giant	Frozen	1,137,267,169	0.069	19 / 21 / 31
Giant	Partial FT	1,137,267,169	10.036	43 / 42 / 49
Giant	LoRA	1,138,741,729	0.199	32 / 36 / 27
Large	Frozen, BCE only	305,023,969	0.215	21 / 26 / 28
Large	Full FT, random init.	305,023,969	100.000	25 / 43 / 26

Table S2. Mean test-volume Dice for each training seed under the main protocol. Controls and the nnU-Net reference are separated.

Backbone	Adaptation	Seed 42	Seed 2026	Seed 3407
Large	Frozen	0.8775	0.8738	0.8727
Large	Partial FT	0.8842	0.8837	0.8875
Large	LoRA	0.8966	0.9010	0.9001
Large	Full FT	0.9052	0.9110	0.9036
Giant	Frozen	0.8777	0.8771	0.8712
Giant	Partial FT	0.8917	0.8894	0.8921
Giant	LoRA	0.9032	0.9023	0.9026
Large	Frozen, BCE only	0.8793	0.8807	0.8829
Large	Full FT, random init.	0.7629	0.7878	0.7410
nnU-Net	ResEncM 3D	0.9306	0.9304	0.9304

Fixed-workload efficiency measurements

We additionally measure E1–E7 sequentially on an RTX 4090, using batch size 4, 448×448 inputs, FP16 automatic mixed precision, four CPU threads, and zero data-loader workers. Each configuration receives 20 warm-up batches followed by three repetitions of the same 100 batches (400 training slices). A separate untimed pass reads the measured slices before timing. The encoder is initialized from pretrained weights and the decoder/adaptation parameters are newly initialized; these are timing runs, not retrained segmentation results.

CUDA synchronization brackets each timed repetition. Timing includes data loading, transfer, forward/loss/backward, optimizer processing, and training metrics, but excludes validation, checkpoint I/O, and test inference. Peak allocated memory is reset per repetition; the table reports the largest of the three allocated peaks in decimal GB. Time and throughput are arithmetic means of the repetition-level measurements. The three repetitions do not estimate independent-seed variability. The launcher checks for other GPU processes before each configuration; this check alone does not certify exclusive access throughout every measurement.

Table S3. Post-hoc exclusion sensitivity. Values are mean ± sample SD over the three seed-wise volume means.

Backbone	Adaptation	All 31 volumes	Remaining 30 volumes
Large	Frozen	0.8747 ± 0.0025	0.8728 ± 0.0026
Large	Partial FT	0.8851 ± 0.0021	0.8835 ± 0.0022
Large	LoRA	0.8992 ± 0.0023	0.8980 ± 0.0023
Large	Full FT	0.9066 ± 0.0039	0.9049 ± 0.0038
Giant	Frozen	0.8753 ± 0.0036	0.8741 ± 0.0035
Giant	Partial FT	0.8911 ± 0.0015	0.8887 ± 0.0014
Giant	LoRA	0.9027 ± 0.0005	0.9015 ± 0.0004
Large	Frozen, BCE only	0.8810 ± 0.0018	0.8786 ± 0.0017
Large	Full FT, random init.	0.7639 ± 0.0234	0.7582 ± 0.0238
nnU-Net	ResEncM 3D	0.9305 ± 0.0001	0.9293 ± 0.0002

Table S4. Robustness of key contrasts (percentage points). Case bootstrap: the analysis of the main text. Case × seed bootstrap: cases and, independently for each configuration, the three seeds are resampled (20,000 draws), so the interval also reflects retraining variation. Least similar: the 21 test volumes whose maximum image correlation with any training volume (split audit, coarse grid) is at most 0.658, a screen for near-duplicate leakage. r: correlation between this similarity and the per-case difference.

Contrast	Case bootstrap	Case × seed bootstrap	Least similar 21	r
DINOv2 init. − random init. (Large Full FT)	14.27 [9.52, 20.38]	[9.14, 21.13]	16.56 [10.11, 24.92]	−0.37
Large: Full FT − Frozen	3.19 [2.73, 3.71]	[2.56, 3.89]	3.34 [2.77, 3.98]	−0.19
Large: LoRA − Frozen	2.46 [2.15, 2.78]	[1.97, 2.95]	2.58 [2.25, 2.96]	−0.36
Large: Partial FT − Frozen	1.05 [0.81, 1.32]	[0.63, 1.51]	1.09 [0.78, 1.46]	−0.23
LoRA: Giant − Large	0.35 [0.16, 0.54]	[0.06, 0.70]	0.41 [0.17, 0.65]	−0.08
Large Full FT − Giant LoRA	0.39 [0.10, 0.69]	[−0.07, 0.88]	0.35 [0.001, 0.68]	+0.12
nnU-Net − Large Full FT	2.39 [1.94, 2.85]	[1.80, 2.98]	2.39 [1.82, 2.96]	−0.15
Common recipe: Full FT − LoRA	0.000 [−0.19, 0.18]	[−0.26, 0.25]	−0.04 [−0.30, 0.21]	+0.02
Common recipe: LoRA − Partial FT	0.36 [0.12, 0.63]	[0.07, 0.67]	0.42 [0.13, 0.73]	−0.17
Common recipe: Full FT − Frozen	0.83 [0.55, 1.11]	[0.51, 1.19]	0.74 [0.45, 1.01]	+0.05

Table S5. Fixed workload: 100 batches per repetition, three repetitions. E2 is flagged because one optimizer update was skipped across the 300 measured batches; its time is not a fully successful-update-matched comparison.

ID	Configuration	Seconds/100 batches	Slices/s	Peak GB
E1	Large Frozen	5.392	74.19	1.609
E2	Large Full FT†	14.117	28.34	8.722
E3	Large Partial FT	6.834	58.53	2.814
E4	Giant LoRA	26.406	15.15	15.888
E5	Giant Frozen	12.538	31.90	4.997
E6	Large LoRA	9.577	41.77	4.854
E7	Giant Partial FT	15.651	25.56	7.309

No measured optimizer updates were skipped for the other six configurations. These measurements characterize the specified short workload, not complete-epoch or full-training duration; multiplying them by historical epoch counts would produce a projection, not an observed runtime.

Experiment identifiers

The identifiers link the manuscript to archived runs: E0 = Large Frozen (BCE only); E1 = Large Frozen; E2 = Large Full FT; E3 = Large Partial FT; E4 = Giant LoRA; E5 = Giant Frozen; E6 = Large LoRA; E7 = Giant Partial FT (blocks 36–39). E1–E7 use the common BCE–Dice objective. E0, E1, E5 and E6 are the reruns with the FP32 loss implementation used by all other configurations; the earlier runs of these four configurations are not used in any reported result.

Frozen-feature objective sensitivity

The BCE-only control (E0) has mean Dice 0.8810, compared with 0.8747 for the combined objective (E1). The paired difference (E0 minus E1) is 0.629 percentage points (95% CI [0.395, 0.867]), using the same seed-averaged case bootstrap as the main analysis. This post-hoc comparison is limited to Large frozen features and changes the BCE coefficient as well as the Dice term.

Validation-trajectory summaries

Skipped optimizer updates over the complete runs (seeds 42/2026/3407): E0: 14/17/18; E1: 27/23/31; E2: 22/31/24; E3: 25/30/18; E4: 30/32/26; E5: 13/20/26; E6: 26/45/23; E7: 40/37/44. These counts are descriptive and are not used to explain the observed declines.

Image-similarity audit

All 11,781 pairs were screened on a $64 \times 64 \times 44$ grid using linear image interpolation and nearest-neighbor mask sampling, without registration. Exploratory screening thresholds were Pearson correlation ≥ 0.95 or mask Dice ≥ 0.90, with exact binary-mask matches also retained. The union with the top 20 pairs by each screening score contained 36 candidates. Of these, 29 shape-matched pairs were compared at native resolution; seven shape-mismatched pairs had screening-grid comparisons only. Native strong-similarity criteria were correlation ≥ 0.99 over the union of nonzero image voxels and mask Dice ≥ 0.98. The identified train–test pair had identical binary masks and image correlation 0.996420. These criteria identify image similarity, not patient identity, and were not prespecified patient-matching rules.

Hardware and software register

Experiments were run on shared institutional GPUs. The register below is transcribed from each run's recorded provenance. Inference of the same checkpoint on an RTX 4090 and an RTX 5090 gave Dice identical to four decimals; retraining the same configuration and seed on the two GPUs differed by 0.40 percentage points (Large LoRA, seed 42: the reported RTX 4090 run and the RTX 5090 run of the single-seed adapter learning-rate sweep, which used PyTorch 2.11.0+cu128 and sources S1 and W1). Two further cross-GPU pairs, which also differ in source snapshot, differed by 0.38 (Large Frozen) and 0.60 points (Large LoRA; seed 3407, main-table run versus last-layer readout cell). These differences are of the order of seed-to-seed variation. Effects of this size or smaller are interpreted with caution. A seed does not guarantee bitwise determinism across hardware or software. S1 and S2 are hashes of the DINOv2 Python source

Table S6. Post-hoc validation stability on the full recorded trajectory. MDD (maximum drawdown) is the largest fall of validation Dice below an earlier value; BTF (best-to-final deterioration) is the best minus the final validation Dice. Both are Dice differences on the 0–1 scale. Best epoch is derived from the validation history. Observation lengths differ.

Configuration	Seed	Epochs	Best epoch	MDD	BTF
Large Frozen	42	31	21	0.0753	0.0257
Large Frozen	2026	26	16	0.0361	0.0207
Large Frozen	3407	34	24	0.0134	0.0123
Large Partial FT	42	25	15	0.1005	0.0125
Large Partial FT	2026	32	22	0.0294	0.0139
Large Partial FT	3407	16	6	0.1620	0.0065
Large LoRA	42	25	15	0.0181	0.0117
Large LoRA	2026	50	46	0.0243	0.0000
Large LoRA	3407	22	12	0.0287	0.0020
Large Full FT	42	18	8	0.8921	0.8908
Large Full FT	2026	28	18	0.5111	0.0544
Large Full FT	3407	21	11	0.8899	0.0093
Giant Frozen	42	19	9	0.0206	0.0031
Giant Frozen	2026	21	11	0.0469	0.0359
Giant Frozen	3407	31	21	0.0718	0.0007
Giant Partial FT	42	43	33	0.0347	0.0048
Giant Partial FT	2026	42	32	0.2164	0.0049
Giant Partial FT	3407	49	39	0.0212	0.0024
Giant LoRA	42	32	22	0.0182	0.0034
Giant LoRA	2026	36	26	0.0236	0.0048
Giant LoRA	3407	27	17	0.0582	0.0085
Large Frozen (BCE)	42	21	11	0.0141	0.0105
Large Frozen (BCE)	2026	26	16	0.0405	0.0012
Large Frozen (BCE)	3407	28	18	0.0070	0.0067
Large Full FT, random init.	42	25	15	0.2389	0.0115
Large Full FT, random init.	2026	43	33	0.7295	0.0833
Large Full FT, random init.	3407	26	16	0.0955	0.0128
Random init., LR 10^{-4} + clip	42	38	28	0.2873	0.0047
Random init., LR 10^{-4} + clip	2026	50	44	0.0994	0.0306
Random init., LR 10^{-4} + clip	3407	50	43	0.0624	0.0137
Large Full FT + clipping	42	24	14	0.0163	0.0163
Large Full FT + clipping	2026	19	9	0.0244	0.0128
Large Full FT + clipping	3407	26	16	0.0352	0.0005
Large Full FT, unclipped rerun	42	21	11	0.6257	0.0126
Common improved: Frozen	42	45	35	0.0145	0.0138
Common improved: Frozen	2026	28	18	0.0164	0.0023
Common improved: Frozen	3407	47	37	0.0102	0.0050
Common improved: Partial FT	42	27	17	0.0105	0.0013
Common improved: Partial FT	2026	33	23	0.0116	0.0059
Common improved: Partial FT	3407	29	19	0.0143	0.0044
Common improved: LoRA	42	34	24	0.0078	0.0038
Common improved: LoRA	2026	26	16	0.0100	0.0031
Common improved: LoRA	3407	35	25	0.0160	0.0019
Common improved: Full FT	42	26	16	0.0123	0.0017
Common improved: Full FT	2026	15	5	0.0248	0.0151
Common improved: Full FT	3407	15	5	0.0108	0.0104

tree; S3 was recorded by the readout runner as a digest of per-file hashes and is not directly comparable with S1 and S2. The nnU-Net reference was trained on an RTX 4090 with PyTorch 2.11.0+cu126 and nnunetv2 2.8.1 for all three seeds; the label-budget nnU-Net runs and the 2D configuration used an RTX 5090 with PyTorch 2.11.0+cu128 and the same nnunetv2 version.

Table S7. Recorded hardware and software for every reported training run. S: DINOv2 source identifier; W: pretrained-weight hash (identifiers listed below the table).

Configuration	Seeds	GPU	PyTorch	Source	Weights
Large Frozen (BCE)	42, 2026, 3407	RTX 4090	2.11.0+cu126	S1	W1
Large Frozen	42, 2026, 3407	RTX 4090	2.11.0+cu126	S1	W1
Large Partial FT	42, 2026, 3407	RTX 4090	2.11.0+cu128	S1	W1
Large LoRA	42, 2026, 3407	RTX 4090	2.11.0+cu126	S1	W1
Large Full FT	42, 2026, 3407	RTX PRO 6000 (MIG 2g.48gb)	2.8.0+cu128	S2	W1
Giant Frozen	42, 2026, 3407	RTX 4090	2.11.0+cu126	S1	W2
Giant Partial FT	42, 2026, 3407	RTX 4090	2.11.0+cu126	S1	W2
Giant LoRA	42, 2026, 3407	RTX 4000 Ada	2.11.0+cu128	S1	W2
Large Full FT, random init.	42, 2026, 3407	RTX 4090	2.11.0+cu126	S1	W1
Random init., LR 10^{-4} + clip	42, 2026, 3407	RTX 5090	2.11.0+cu128	S1	W1
Large LoRA, LR 10^{-4}	42, 2026, 3407	RTX 4090	2.11.0+cu126	S1	W1
Giant LoRA, LR 10^{-4}	42, 2026, 3407	RTX 4090	2.11.0+cu126	S1	W2
Large Full FT + clipping	42, 2026, 3407	RTX 5090	2.11.0+cu128	S1	W1
Large Full FT, unclipped rerun	42	RTX 5090	2.11.0+cu128	S1	W1
Readout: Frozen, last layer	42, 2026	RTX 4090	2.11.0+cu126	S3	W1
Readout: Frozen, last layer	3407	RTX 5090	2.11.0+cu128	S3	W1
Readout: Frozen, four layers	42, 2026	RTX 4090	2.11.0+cu126	S3	W1
Readout: Frozen, four layers	3407	RTX 5090	2.11.0+cu128	S3	W1
Readout: LoRA, last layer	42, 2026	RTX 4090	2.11.0+cu126	S3	W1
Readout: LoRA, last layer	3407	RTX 5090	2.11.0+cu128	S3	W1
Readout: LoRA, four layers	42, 2026	RTX 4090	2.11.0+cu126	S3	W1
Readout: LoRA, four layers	3407	RTX 5090	2.11.0+cu128	S3	W1
Common improved: Large Frozen	42, 2026, 3407	RTX 5090	2.11.0+cu128	S3	W1
Common improved: Large Partial FT	42, 2026, 3407	RTX 5090	2.11.0+cu128	S3	W1
Common improved: Large LoRA	42, 2026, 3407	RTX 5090	2.11.0+cu128	S3	W1
Common improved: Large Full FT	42, 2026, 3407	RTX 5090	2.11.0+cu128	S3	W1
Label budget, all 34 runs	42, 2026, 3407	RTX 5090	2.11.0+cu128	S3	W1
Decoder study, all 18 runs	42, 2026, 3407	RTX 5090	2.11.0+cu128	S3	W1

S1: `python-tree-sha256:041ad5d1e3d0d97986db3d8c8db60a8723006ea1250f7c112ddbdc178c634bf6`

S2: `python-tree-sha256:ec0ecd3faeceefdc22111637160ec105892c1d0b6742768eb5014b56ee6b879c`

S3: `file-sha256:e446be121dc361d4751953c4d393a7e19ee8e5e0759c83e309550b5f86743cca`

W1: `d5383ea8f4877b2472eb973e0fd72d557c7da5d3611bd527ceeb1d7162cbf428`

W2: `baf8467e50af277596bbbafa06887c177ee899ab46033649c383577d7e9309d3`

Table S12 lists when each analysis was fixed relative to the test results. All analyses score the same 31 test volumes, and none has an untouched confirmation set.

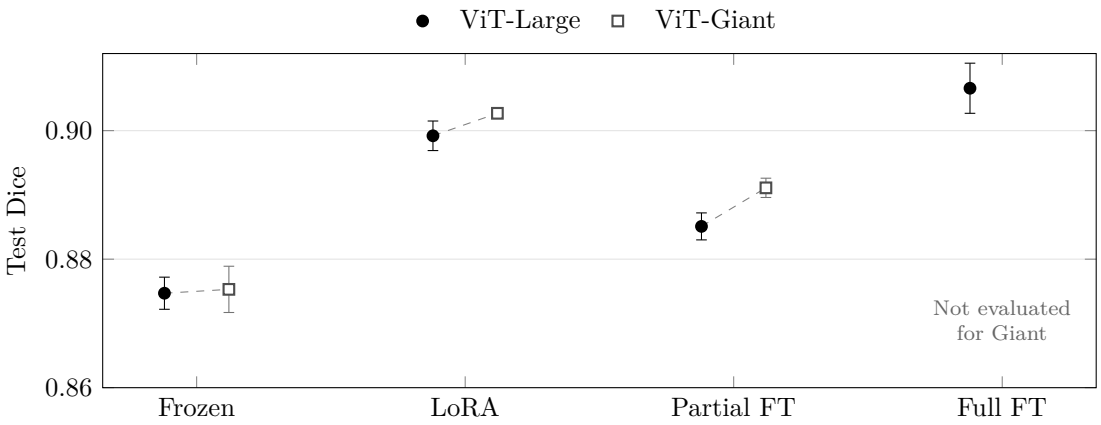


Figure S1. Observed strategy–scale matrix under the main protocol. Markers are three-seed mean Dice and error bars are sample SD across seeds, not confidence intervals. Dashed lines connect matched Large–Giant comparisons. Giant Partial FT uses the last four blocks, matching block count rather than depth fraction. Giant Full FT was not evaluated; paired CIs are given in Table 4.

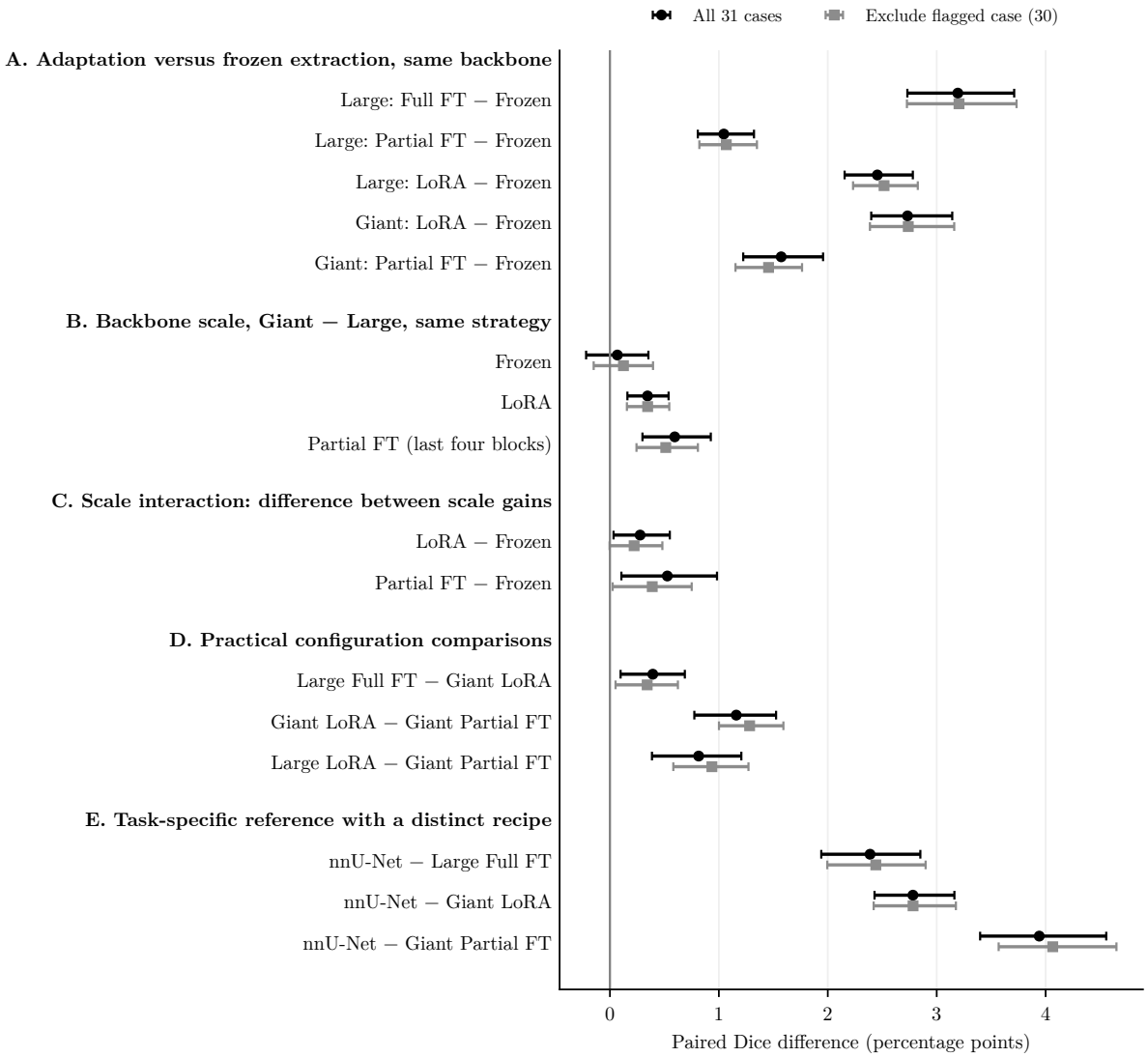


Figure S2. Paired Dice contrasts of Table 4, in the same groups, for all 31 cases (black circles) and the 30-case exclusion sensitivity (grey squares). Lines denote 95% percentile case-bootstrap CIs; the vertical reference is zero. These intervals do not include retraining variance or certify patient independence.

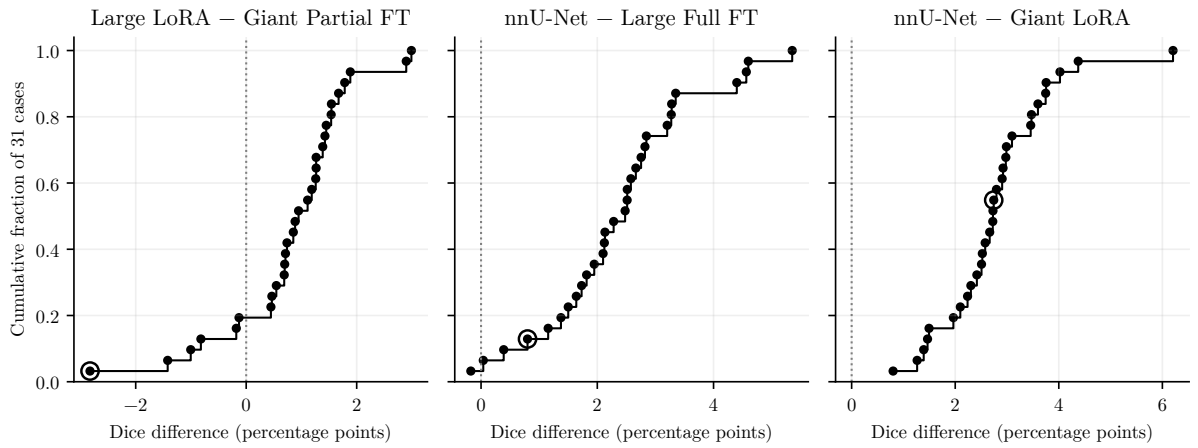


Figure S3. Empirical cumulative distributions of seed-averaged paired Dice differences for all 31 cases. Every marker is one case; the outlined marker identifies the flagged near-duplicate. Positive values favor the first configuration.

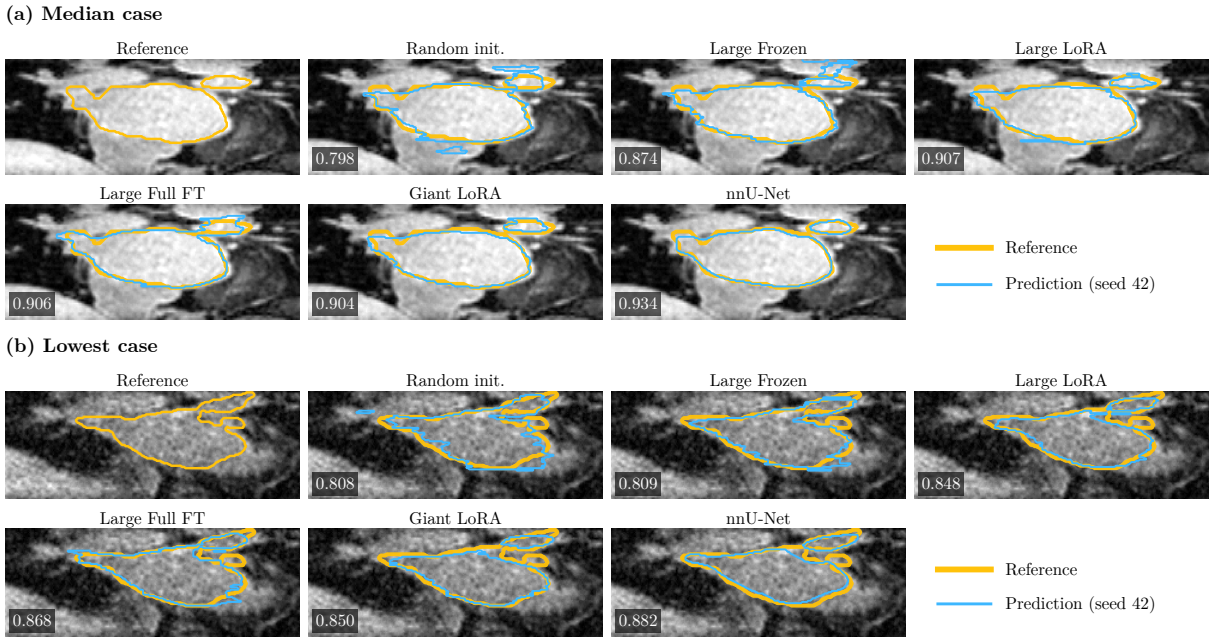


Figure S4. Coronal reformats (front view; superior at top, patient left on the right) of the two volumes of Figure 2, (a) median-ranked, (b) lowest-ranked, through the coronal plane with the largest reference LA area; each panel is centred on the LA and the predictions. The volumes cover only the atrial region in the slice direction, so the panels are short. Yellow (thicker, drawn below): reference; blue: prediction (seed 42); numbers are volume-level Dice. Intensity windowing (1st–99.5th percentile) changes display only.

Table S8. Mean test-volume Dice for each training seed in the decoder study (ViT-Large, common improved recipe, hierarchical decoder) and for nnU-Net ResEncM 2D, trained with the recipe of the 3D reference on the same 108 training volumes. Selected epoch: epoch of the retained checkpoint for seeds 42/2026/3407 (DINOv2: highest validation Dice within at most 50 epochs and patience 10; nnU-Net: last improvement of the moving-average validation-patch Dice within 1,000 epochs).

Training volumes	Configuration	Seed 42	Seed 2026	Seed 3407	Selected epoch
108	Frozen	0.9071	0.9075	0.9099	25/21/20
108	Partial FT	0.9115	0.9108	0.9054	33/10/5
108	LoRA	0.9137	0.9100	0.9136	10/5/10
108	Full FT	0.9163	0.9129	0.9153	15/5/8
11	LoRA	0.8576	0.8804	0.8911	4/14/8
11	Full FT	0.8461	0.8519	0.8626	1/4/3
108	nnU-Net ResEncM 2D	0.9238	0.9231	0.9229	980/728/350

Table S9. Recipe sensitivity within each strategy (RQ3; post hoc, three seeds, selection on validation only). Each factor was varied for the strategy shown; rows are not compared across strategies. Δ is the paired case-bootstrap difference (percentage points, unadjusted 95% CI) against the reference stated in the row; per-seed differences follow seed order 42/2026/3407.

Factor (strategy)	Reference	Variant	Δ [95% CI]	Cases > 0	Per seed
Adapter LR 10^{-4} (Large LoRA)	0.8992 ± 0.0023	0.9038 ± 0.0009	0.46 [0.20, 0.75]	24/31	0.63/0.38/0.36
Adapter LR 10^{-4} (Giant LoRA)	0.9027 ± 0.0005	0.9053 ± 0.0007	0.26 [0.001, 0.51]	23/31	0.28/0.31/0.20
Giant − Large at LR 10^{-4} (LoRA)	0.9038 ± 0.0009	0.9053 ± 0.0007	0.15 [−0.004, 0.30]	21/31	0.30/0.06/0.09
Gradient clipping 1.0 (Large Full FT)	0.9066 ± 0.0039	0.9125 ± 0.0006	0.59 [0.41, 0.77]	27/31	0.67/0.18/0.92
Four-layer readout (Large Frozen)	0.8757 ± 0.0007	0.8981 ± 0.0012	2.24 [1.80, 2.67]	30/31	2.20/2.22/2.29
Four-layer readout (Large LoRA)	0.8971 ± 0.0027	0.9044 ± 0.0022	0.73 [0.47, 0.98]	28/31	0.51/0.87/0.81

The adapter learning rate was selected on validation in a single-seed sweep of Large LoRA (10^{-5}, 10^{-4}; 10^{-3} diverged) and applied unchanged to Giant LoRA. Clipped Full FT runs used an RTX 5090 and the reference runs an RTX PRO 6000; Giant LoRA reference runs used an RTX 4000 Ada and its 10^{-4} runs an RTX 4090. An unclipped seed-42 rerun on the RTX 5090 reached Dice 0.9064, against 0.9119 with clipping (its validation MDD is in Table S6). Readout cells are independent retrainings with the main protocol otherwise unchanged; seeds 42 and 2026 used an RTX 4090 and seed 3407 an RTX 5090, with the four cells of a seed on one GPU. The four-layer readout partitions the original projection and adds no trainable parameters.

Table S10. Reduced label budgets (RQ1; exploratory, design fixed before any reduced-data run but after the main results). ViT-Large configurations use the common improved recipe without augmentation and nnU-Net its own recipe with augmentation; each seed trains on one nested random subset of the 108 training volumes, shared by all configurations, and every run selects its checkpoint on the same 15 annotated validation volumes and is tested on the same 31 volumes. Random init.: the same ViT-Large trained from random initialization with the common recipe and an encoder learning rate of 10^{-4}; its 108-volume runs are new and therefore differ from both from-scratch rows of Table 3. The other 108-volume DINOv2 rows are the runs of Table 5. Upper block: Dice, mean ± SD of three seed means. Lower block: paired case-bootstrap differences in percentage points with 95% CIs; the last column is the change of each quantity from 108 to 11 training volumes.

Training volumes	11 (10%)	27 (25%)	108 (100%)	Change, 11 − 108
Dice				
Random init. (Full FT)	0.4615 ± 0.0386	0.6260 ± 0.0374	0.7616 ± 0.0070	−30.01 [−34.40, −25.91]
Frozen	0.8611 ± 0.0088	0.8837 ± 0.0018	0.9032 ± 0.0008	−4.22 [−5.49, −3.11]
Partial FT	0.8724 ± 0.0064	0.8859 ± 0.0007	0.9079 ± 0.0010	−3.56 [−4.38, −2.81]
LoRA	0.8832 ± 0.0059	0.8981 ± 0.0012	0.9115 ± 0.0007	−2.84 [−3.54, −2.18]
Full FT	0.8810 ± 0.0056	0.8945 ± 0.0013	0.9115 ± 0.0011	−3.05 [−3.65, −2.50]
nnU-Net ResEncM 3D	0.9203 ± 0.0008	0.9246 ± 0.0017	0.9305 ± 0.0001	−1.02 [−1.44, −0.65]
Paired differences, percentage points				
Full FT − random init.	41.95 [36.55, 47.66]	26.85 [22.17, 32.30]	14.99 [10.74, 20.46]	26.96 [22.99, 31.17]
LoRA − Full FT	0.22 [−0.17, 0.58]	0.37 [−0.24, 1.22]	0.000 [−0.18, 0.19]	0.22 [−0.19, 0.58]
LoRA − Frozen	2.21 [1.40, 3.19]	1.45 [0.97, 2.05]	0.83 [0.55, 1.11]	1.38 [0.64, 2.25]
nnU-Net − LoRA	3.71 [3.05, 4.42]	2.65 [2.23, 3.10]	1.90 [1.54, 2.26]	1.82 [1.30, 2.39]
nnU-Net − Full FT	3.93 [3.31, 4.62]	3.01 [2.24, 4.07]	1.90 [1.49, 2.32]	2.03 [1.56, 2.54]

p_{Holm} over the 15 contrasts at 11 and 27 volumes and in the change column (family defined after these results were viewed): LoRA − Full FT, 11 volumes 0.782, 27 volumes 0.782, change 0.782; all others < 0.002. The 108-volume contrasts carry unadjusted intervals. Every contrast whose interval excludes zero has the same sign in all three seeds, and a case × seed bootstrap that also resamples the subsets excludes zero for the same contrasts and only for them. Every reduced-data epoch draws 2,376 batches, as a full-data epoch does, so all budgets share the update schedule and early-stopping rule. nnU-Net plans were derived from the 123 development volumes for every budget.

Table S11. Augmentation study (post hoc; design and nine contrasts fixed before any augmented run). The common improved recipe of the label-budget analysis, with nnU-Net-style 2D augmentation of the training slices (rotation, scaling, noise, blur, brightness, contrast, low resolution, gamma, mirroring) as the only change; same nested subsets, 50-epoch limit, early stopping, validation and test volumes. Upper block: Dice, mean ± SD of three seed means. Lower block: paired case-bootstrap differences in percentage points with 95% CIs and p_{Holm} over the nine contrasts.

	11 training volumes	108 training volumes	Change, 11 − 108
Dice			
LoRA, augmented	0.8846 ± 0.0068	0.9075 ± 0.0041	
LoRA	0.8832 ± 0.0059	0.9115 ± 0.0007	
Random init., augmented	0.5099 ± 0.0902	0.6928 ± 0.0142	
Random init.	0.4615 ± 0.0386	0.7616 ± 0.0070	
nnU-Net ResEncM 3D	0.9203 ± 0.0008	0.9305 ± 0.0001	
Paired differences, percentage points (p_{Holm})			
Pretraining gain, augmented (LoRA − random init.)	37.48 [31.44, 44.16] (< 0.002)	21.46 [17.19, 26.49] (< 0.002)	16.01 [12.49, 20.03] (< 0.002)
Augmentation effect, random init.	4.84 [1.18, 8.54] (0.017)	−6.88 [−10.33, −3.84] (< 0.002)	
Augmentation effect, LoRA	0.15 [−0.29, 0.56] (0.492)	−0.41 [−0.74, −0.11] (0.016)	
nnU-Net − augmented LoRA	3.57 [2.87, 4.33] (< 0.002)	2.30 [1.83, 2.81] (< 0.002)	

Epochs completed/selected epoch (seeds 42/2026/3407): LoRA, 108 volumes 35/25, 17/7, 31/21; LoRA, 11 volumes 31/21, 16/6, 30/20; random init., 108 volumes 50/48, 50/45, 50/49; random init., 11 volumes 18/8, 21/11, 50/44. 4 of the six augmented random-init. runs reached the 50-epoch limit, so they had not converged. Unaugmented references: label-budget runs, and the common-recipe LoRA runs at 108 volumes.

Table S12. Status of the analyses. "Fixed" states when the configurations and contrasts were defined; "unadjusted" means 95% intervals without multiplicity correction. Run counts exclude superseded runs (Methods).

Analysis	Runs	Fixed	Multiplicity
Main protocol (RQ1, RQ2)	8 DINOv2 configurations and 2 from-scratch recipes, 3 seeds (30)	Before any run; the 16-contrast family after the main test results	Holm (16)
Recipe factors (RQ3)	5 single-factor variants, 3 seeds (15), and 1 unclipped rerun	After the main test results; selection on validation data only	Unadjusted
Common improved recipe (RQ3)	4 ViT-Large strategies, 3 seeds (12)	After the main test results; test scored only after all 12 runs	Unadjusted
Label budget (RQ1)	34 ViT-Large runs and 6 nnU-Net runs	After the main and common-recipe test results, before any reduced-data run	Holm (15)
Decoder capacity (RQ3)	18 runs	After the common-recipe and label-budget results, before these runs	Holm (12)
nnU-Net 2D	3 runs	With the decoder study, after the main results	Unadjusted
Augmentation (RQ1)	LoRA and random init., 108 and 11 volumes, 3 seeds (12)	After all other results, before any augmented run	Holm (9)
External evaluation (RQ4)	Main-protocol and nnU-Net checkpoints; common-recipe checkpoints	Main-protocol scoring planned; common-recipe scoring and failure-volume exclusions post hoc	Unadjusted

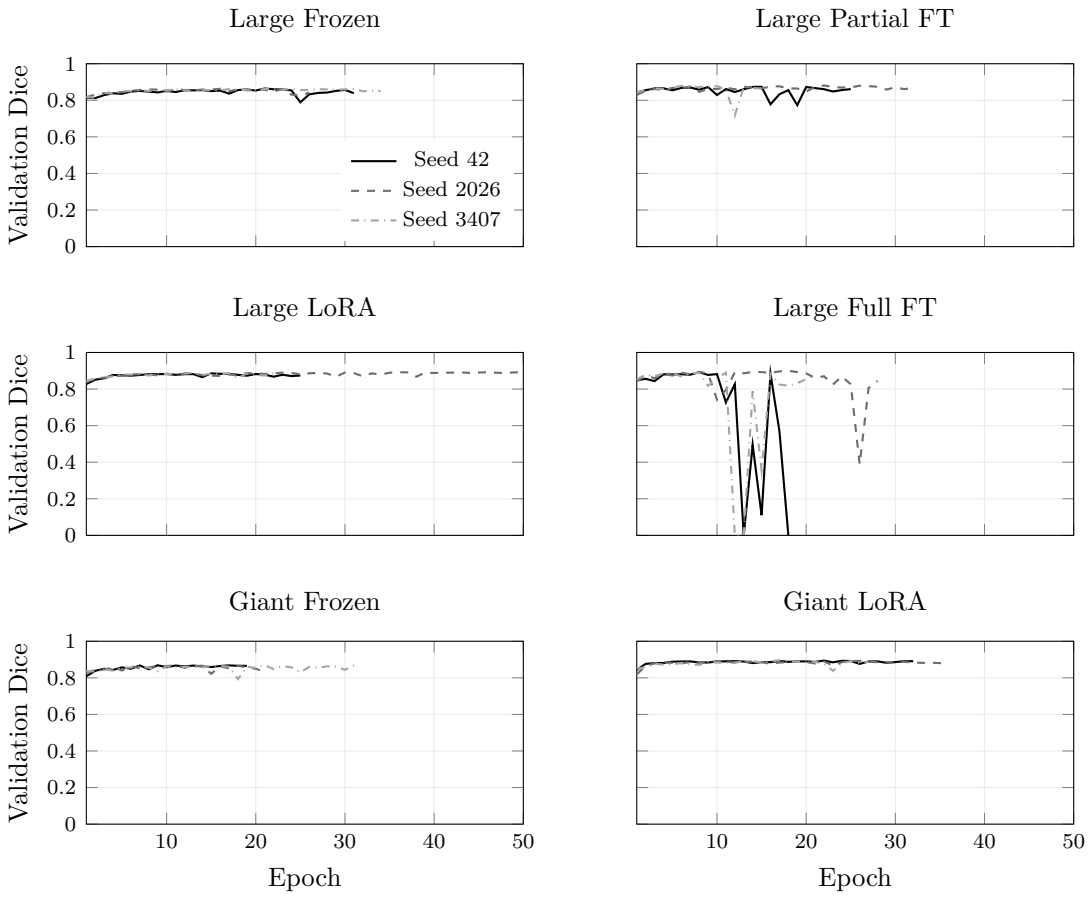


Figure S5. Recorded validation-volume Dice for the six main-protocol configurations other than Giant Partial FT, whose trajectories appear in Figure S7. All 18 trajectories (six configurations, three seeds) are shown over the full Dice range. Black solid, dark-grey dashed and light-grey dash-dotted lines denote seeds 42, 2026 and 3407, respectively. Figure S6 shows the same trajectories with a zoomed axis and maximum-validation epoch markers; it is not an additional experiment. Curves stop at the recorded final epoch; no values are extrapolated after early stopping. These are validation trajectories, not test results.

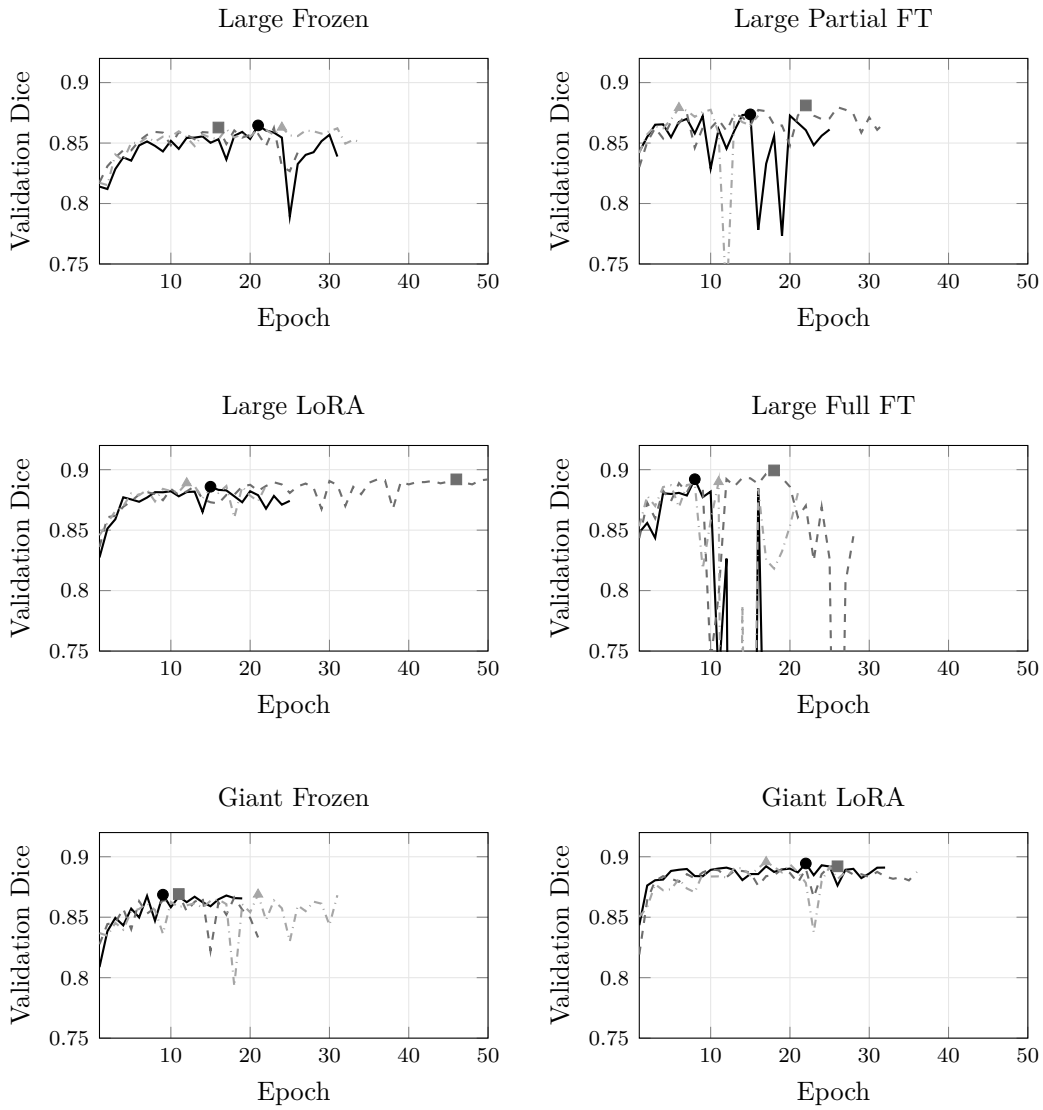


Figure S6. Zoomed validation trajectories (0.75–0.92) with the maximum-validation epoch from each recorded history marked for each seed. Markers do not independently verify checkpoint contents. Black solid with circles: 42; dark-grey dashed with squares: 2026; light-grey dash-dotted with triangles: 3407. Values outside the displayed range are clipped here; the complete trajectories, including all collapses, remain in Figure S5.

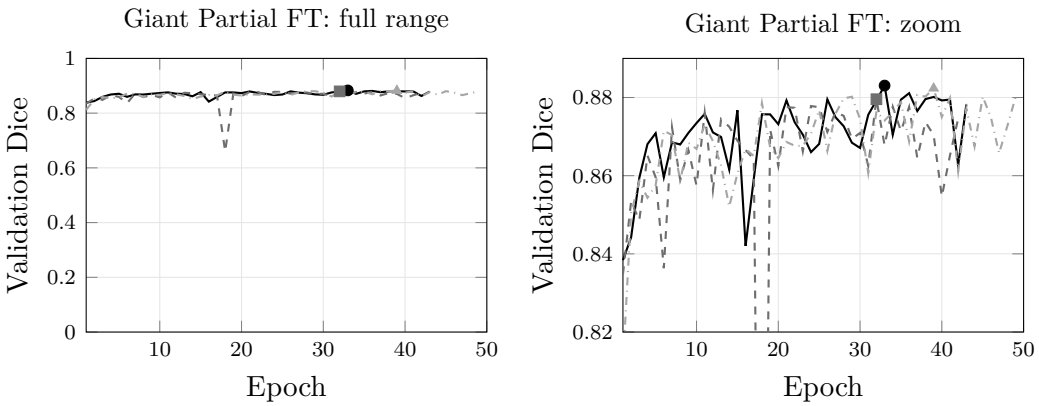


Figure S7. Giant Partial FT validation trajectories from all three completed runs. Black solid with circles: seed 42; dark-grey dashed with squares: 2026; light-grey dash-dotted with triangles: 3407. Dots mark validation-selected epochs 33, 32, and 39. Left: full 0–1 range; right: zoomed view of the same unsmoothed records.